

NGUYEN HOANG HAI

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CAREER OBJECTIVE

Computer Science student at VNU-HUS focusing on Machine Learning and Natural Language Processing. Experienced in building AI applications involving RAG systems, intent classification, and LLM-based solutions. Currently applying advanced frameworks like LangGraph in a real-world internship at EMS, seeking to continuously contribute to and grow within impactful AI product development.

EDUCATION

VNU University of Science (HUS)

Bachelor of Computer Science

2023 – Expected 2027

Hanoi, Vietnam

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Systems, Artificial Intelligence, Machine Learning, Natural Language Processing.
- **Languages:** Vietnamese (Native), English (B2 - capable of reading technical documentation and AI research papers).

TECHNICAL SKILLS

Languages: Python, Java, SQL.

AI/ML: PyTorch, Scikit-learn, Hugging Face Transformers, Sentence-Transformers, LangChain, LangGraph.

Generative AI: RAG Systems, Hybrid Search, Vector Databases, LLM Inference, Intent Classification.

Databases: PostgreSQL, ChromaDB (Vector DB), Redis.

Tools & Concepts: Git, Docker, Flask, OOP, Data Structures & Algorithms.

WORK EXPERIENCE

EMS

AI Engineer Intern

May 2026 – Present

Hanoi, Vietnam

Customer Support AI Chatbot (RAG, Hybrid Search, LangGraph):

- Engineered a scalable customer-facing chatbot across multiple Fanpages, utilizing **LangGraph** for complex conversation routing, state management, and fallback handling.
- Optimized conversational UX and system performance by implementing location-based branch matching, **Redis** message queuing, and robust retry mechanisms.
- Integrated a Telegram alert system for real-time monitoring of failed or paused conversational flows.

Instructor Payroll Automation System:

- Designed and deployed an end-to-end automated pipeline for instructor payroll, handling file ingestion, business logic validation, and dynamic Excel report generation via **Google Sheets** integration.
- Developed a Telegram Admin Bot enabling secure, real-time CRUD operations on instructor profiles and payroll data management.

HIGHLIGHTED PROJECTS

1. Medical QA Chatbot (RAG System) — *Independent Project*

- **Technologies:** Python, LangChain, ChromaDB, Hugging Face, PyTorch, Llama.cpp (Qwen2.5-7B).
- Developed an end-to-end RAG medical chatbot using **Llama.cpp (Qwen2.5-7B)** to provide safe, context-aware answers based on Tam Anh Hospital data.
- Built a semantic retrieval pipeline with **ChromaDB** and PhoBERT, integrating a Cross-Encoder re-ranker to significantly boost document extraction accuracy.
- Implemented advanced chunking strategies and strict prompt engineering to prevent AI hallucination in critical medical topics.

2. Personalized AI Voice Assistant — [GitHub] — *Cross-functional Team Project*

- **Technologies:** Python, Scikit-learn, Flask, PostgreSQL, Sentence-Transformers.

- Architected 4 core NLP-driven modules (QA, Calendar, Personalization, Web Automation) for a scalable voice assistant integrating GPT and **Whisper STT**.
- Engineered a Machine Learning pipeline (**TF-IDF, Logistic Regression**) for Vietnamese intent classification, ensuring accurate user text processing and system stability.